

10.3.25

EE25BTECH11022 - sankeerthan

Question: Find equation of all lines having slope 2 which are tangents to curve $y + \frac{2}{x-3} = 0$

Solution: Given curve:

$$y + \frac{2}{x-3} = 0 \implies yx - 3y + 2 = 0 \quad (1)$$

compare with

$$(x)^T \mathbf{V}(x) + 2(u)^T(x) + f = 0 \quad (2)$$

$$\mathbf{V} = \begin{pmatrix} 0 & 0.5 \\ 0.5 & 0 \end{pmatrix}, (u) = \begin{pmatrix} 0 \\ -1.5 \end{pmatrix}, f = 2 \quad (3)$$

From above, the normal vector is

$$(n) = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad (4)$$

Points of contact:

$$(q)_i = \mathbf{V}^{-1}(\kappa_i(n) - (u)), i = 1, 2 \quad (5)$$

$$\mathbf{V}^{-1} = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}, \quad (6)$$

$$(u)^T \mathbf{V}^{-1}(u) = 0, f = 2, (n)^T \mathbf{V}^{-1}(n) = -8 \quad (7)$$

$$\kappa_i = \pm \sqrt{\frac{(u)^T \mathbf{V}^{-1}(u) - f}{(n)^T \mathbf{V}^{-1}(n)}} \quad (8)$$

$$\kappa_i = \pm \sqrt{\frac{-2}{-8}} = \pm 0.5 \quad (9)$$

For $\kappa_1 = 0.5$:

$$(q)_1 = \mathbf{V}^{-1}(0.5(n) - (u)) = \mathbf{V}^{-1} \begin{pmatrix} -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix} \quad (10)$$

For $\kappa_2 = -0.5$:

$$(q)_2 = \mathbf{V}^{-1}(-0.5(n) - (u)) = \mathbf{V}^{-1} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad (11)$$

Tangent equation at contact point:

$$(\mathbf{V}(q) + (u))^T(x) + (u)^T(q) + f = 0 \quad (12)$$

For $(q)_1 = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$:

$$\mathbf{V}(q)_1 + (u) = \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 0 \\ -1.5 \end{pmatrix} = \begin{pmatrix} -1 \\ 0.5 \end{pmatrix} \quad (13)$$

$$(u)^T (q)_1 = 3 \quad (14)$$

For $(q)_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$:

$$\mathbf{V}(q)_2 + (u) = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ -1.5 \end{pmatrix} = \begin{pmatrix} 1 \\ -0.5 \end{pmatrix} (u)^T (q)_2 = -3 \quad (15)$$

The equations of tangents are

$$\begin{pmatrix} -1 & 0.5 \end{pmatrix} \mathbf{x} + 5 = 0 \quad (16)$$

$$\begin{pmatrix} 1 & -0.5 \end{pmatrix} \mathbf{x} - 1 = 0 \quad (17)$$

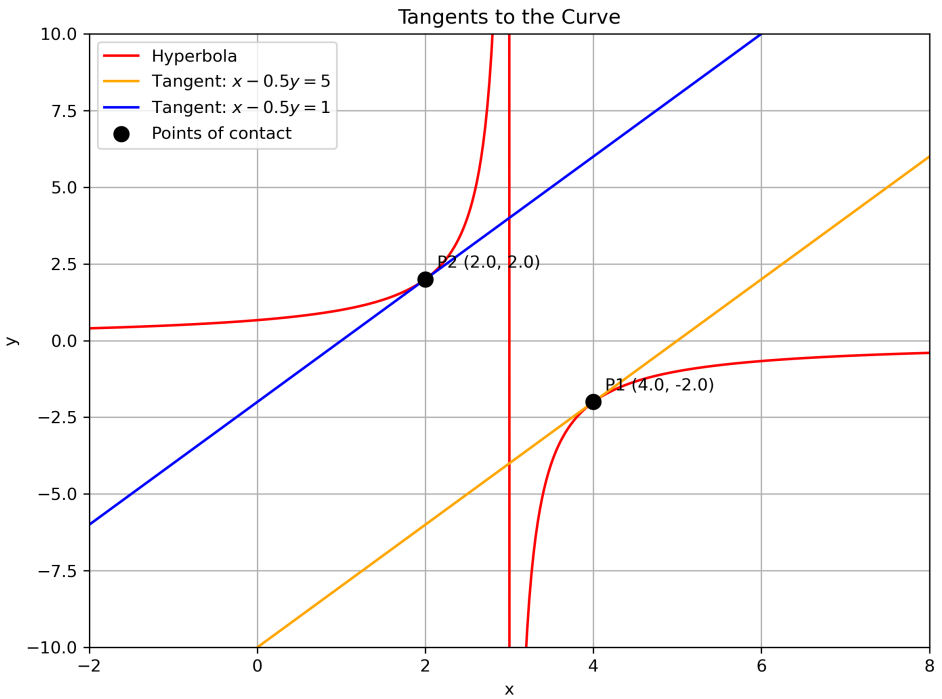


Fig. 0.1